

# CYLINDER DISTRIBUTION PLATFORM.

*Production-Grade LPG Delivery Backend*

## TEAM CYLDIST

Department of Computer Engineering

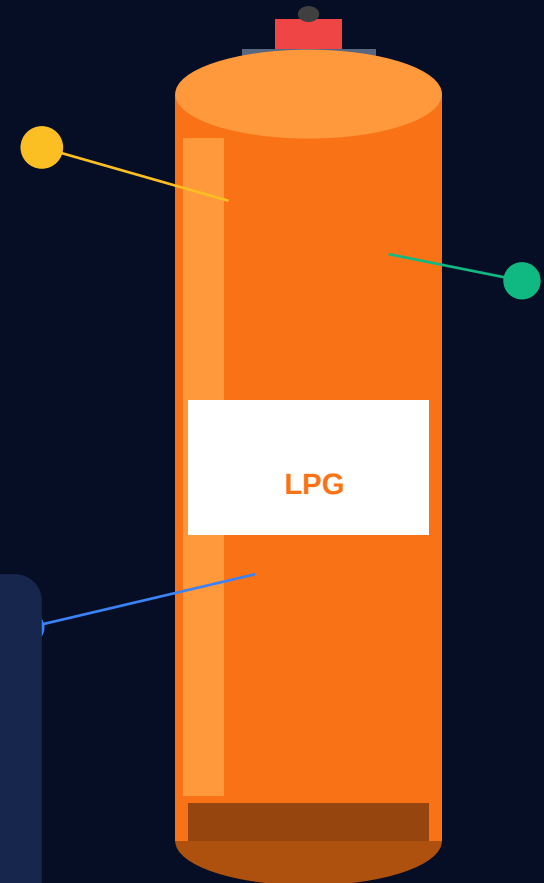
Member 1 - Lead Developer

Member 2 - Backend Engineer

Member 3 - DevOps & DB

Member 4 - Frontend & QA

[Your College Name] | Academic Year 2025-26



# 02

## INTRODUCTION

# Digital LPG Distribution Reimagined.

A production-grade backend that transforms how LPG cylinders are booked, tracked, and delivered.

### PROJECT IMPACT

3+

User Roles

30+

API Endpoints

10+

Core Features

2

Databases



## KEY OBJECTIVES

# Goals & Mission

1

## DIGITIZE

Replace manual booking  
with online + WhatsApp

2

## TRACK

Real-time GPS tracking  
& live customer chat

3

## SECURE

JWT + RBAC + bcrypt  
production security

4

## SCALE

Docker + Kubernetes  
ready deployment

# Problem vs Solution



## TRADITIONAL SYSTEM

Outdated - Manual - Inefficient

-  Manual booking via phone calls
-  No real-time delivery tracking
-  Opaque inventory management
-  Zero priority for emergencies
-  Paper-based record keeping
-  No customer communication
-  Inefficient route planning



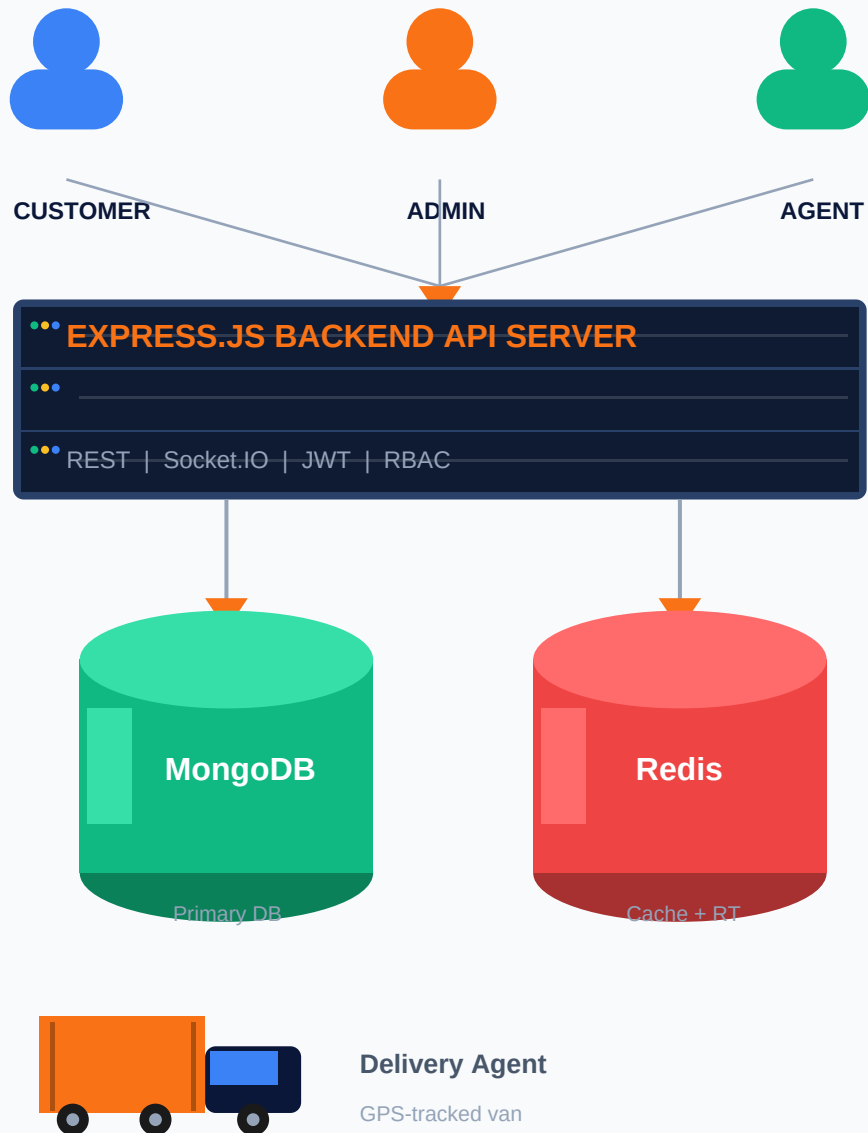
## OUR DIGITAL SOLUTION

Modern - Automated - Scalable

-  Online + WhatsApp booking
-  Live GPS delivery tracking
-  Automated inventory engine
-  Crisis prioritization system
-  Digital order management
-  Real-time agent-customer chat
-  Smart route optimization

# Architecture & Workflow

## ARCHITECTURE FLOW



## ORDER LIFECYCLE FLOW

- 01 Customer places order
- 02 Admin assigns agent
- 03 Agent picks cylinder
- 04 GPS tracking begins
- 05 Live chat enabled
- 06 Delivery + OTP verify
- 07 Order completed!

# Modern Technologies

3D-illustrated stack for production scalability



## SUPPORTING LIBRARIES & TOOLS

**Security:** bcrypt | Helmet | CORS | Zod

**Integration:** Twilio | SendGrid | Razorpay | S3

**Logging:** Winston | Morgan | rotate-file

**Testing:** Jest | Supertest | mongo-mem-server

# Unique Features

01

## WhatsApp Booking

API-based booking via WhatsApp

02

## Product Store

Buy accessories with cylinders

03

## Live GPS Tracking

5-second GPS update intervals

04

## Real-time Chat

Customer-Agent messaging

05

## Crisis Engine

Emergency-aware allocation

06

## Inventory Manager

Warehouse-level cylinder tracking

07

## JWT + RBAC

Role-based access control

08

## Multi-channel Notify

SMS / Email / Push alerts

09

## Swagger API Docs

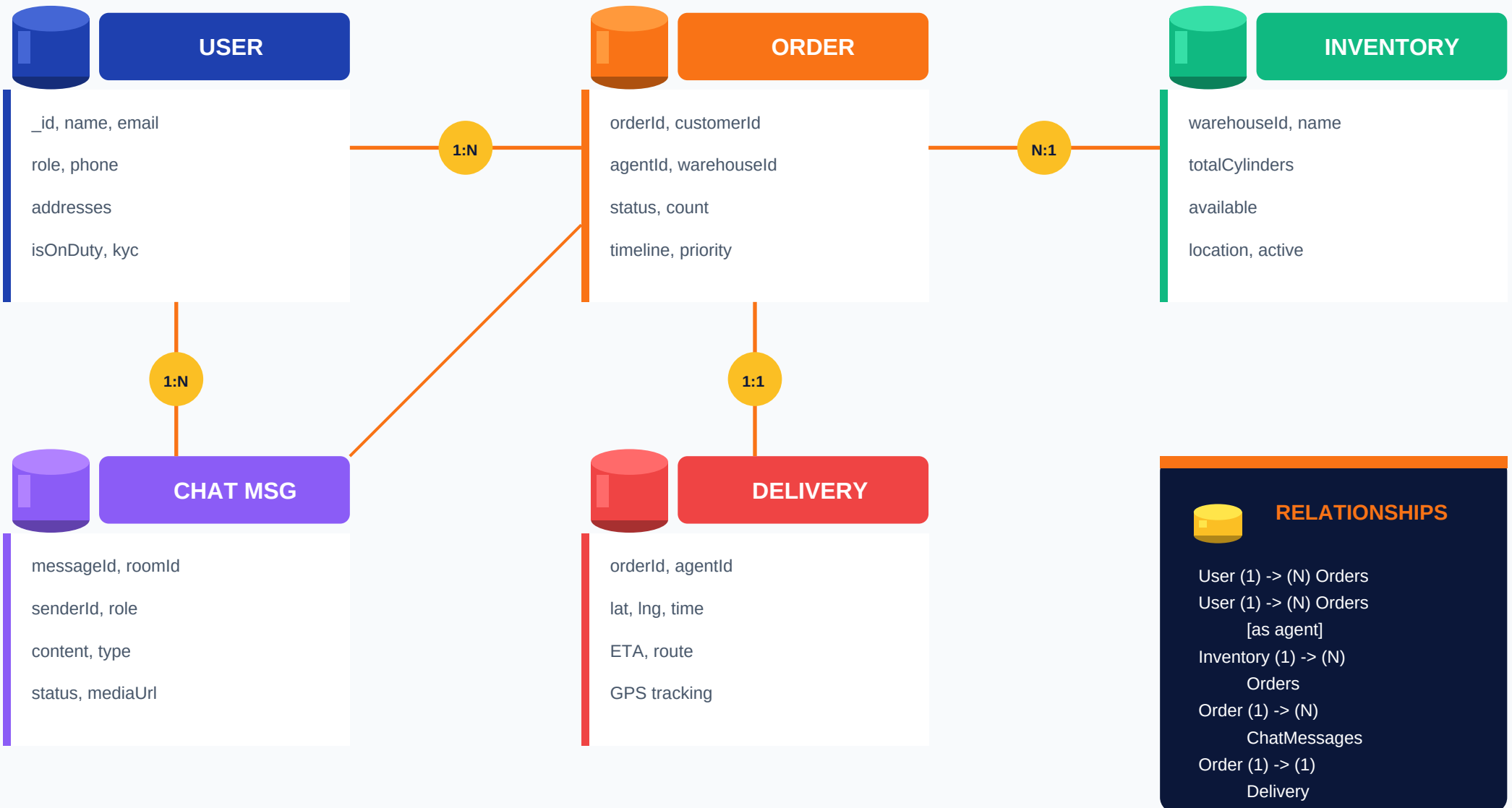
Interactive API testing UI

10

## Docker + K8s

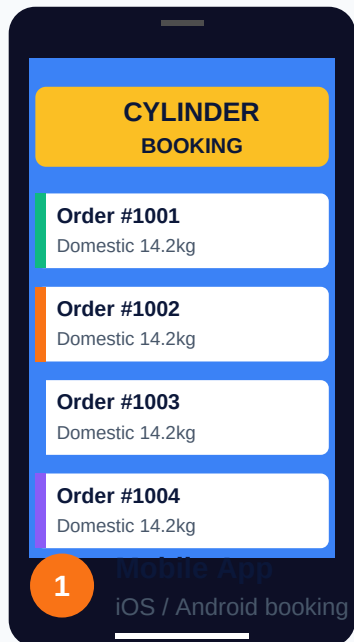
Production-grade deployment

# Database Design & ER Diagram



## Demo &amp; Security

## EXECUTION DEMO



1

Mobile App  
iOS / Android booking

2

**Admin Panel**

Order management UI

3

**Live GPS**

Real-time tracking

4

**API Docs**

Swagger interactive UI



## SECURITY ARCHITECTURE

**MULTI-LAYER PROTECTION**

Production-grade

**JWT AUTH**Access (15min) + Refresh (7d)  
Token rotation + reuse detect**RATE LIMITING**100 req/15min global  
10 req/15min for auth**ENCRYPTION**bcrypt 12-round hashing  
Helmet + CSP + HTTPS**RBAC**Admin | Customer | Agent  
Role-scoped routes

Helmet

CORS

HPP

Sanitize

XSS

Winston

# Conclusion & Future Scope

## PROJECT ACHIEVEMENTS

- ✓ Production-grade REST API
- ✓ Real-time GPS + Live Chat
- ✓ Crisis Prioritization Engine
- ✓ Complete Order Lifecycle
- ✓ Docker & Kubernetes Deploy
- ✓ Multi-layer Security (JWT/RBAC)
- ✓ WhatsApp Integration
- ✓ Swagger API Documentation
- ✓ Inventory Management
- ✓ Real-time Notifications

## FUTURE ENHANCEMENTS

- AI**  
Demand Prediction  
ML-based consumption forecast
- APP**  
Mobile App  
React Native / Flutter
- PAY**  
Online Payments  
Razorpay / Stripe / UPI
- MAP**  
Route Optimization  
Google Maps integration
- IoT**  
Smart Sensors  
Cylinder gas-level IoT

# THANK YOU!

Questions & Discussion Welcome